

Decomposition and structure diagrams

Revise 360

Worksheet for the 360° experience: Decomposition workshop

Name: _____

Class: _____

Date: _____

How to use this sheet

1. Discuss or research the Do Now on slide 1. Complete the starter and key terminology below.
2. Open Decomposition workshop. Read each station panel and write the key facts in your own words. Complete its worksheet tasks before tapping the badge.
3. Answer the questions on screen. Your first answer counts. Use review mode to learn from mistakes.
4. Copy your score into the yellow box, then complete the exam practice from memory.

By the end of the lesson I will:

- Decompose a problem into sub-problems.
- Interpret and create structure diagrams.
- Explain benefits of decomposition.

Starter: before you put the headset on

A school club needs to register members, lend equipment and record returns. Name three smaller problems.

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Key terminology

Decomposition

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Structure diagram

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The stations

Stations 1 and 2: turn right. Stations 3 and 4: turn around. Stations 5 and 6: turn left. Final challenge: look down.

1 Split the problem

Turn right

Key facts in your own words:

Challenge: Suggest three parts of a library system.

Which example is decomposition?

Which is a sub-problem of a loan system?

2 Read the hierarchy

Turn right

Key facts in your own words:

Challenge: What are the two parts of Issue loan?

What does a line to a child box mean?

The top box of a structure diagram is...

3 Refine a sub-problem

Turn around

Key facts in your own words:

Challenge: Break Process return into three smaller tasks.

Order a sensible refinement process.

Put these steps in order: Check the parts cover the requirements / Split a complex sub-problem / Name its main sub-problems / Identify the whole system

Refinement means...

4 Structure versus flow

Turn around

Key facts in your own words:

Challenge: Why is a structure diagram not a flowchart?

Classify the diagram features.

Categories: Structure diagram / Flowchart

Item	Category
Parent and child modules	
Yes/no decision branch	
System broken into parts	
Arrow showing next action	

Does left-to-right position in a structure diagram define execution order?

5 Work as a team

Turn left

Key facts in your own words:

Challenge: Why must both programmers agree on member ID?

Which is a benefit of decomposition?

A shared interface helps parts...

6 Check coverage

Turn left

Key facts in your own words:

Challenge: Which responsibility is missing?

The system must report overdue loans. What should be added?

An omitted return function is found by checking...

★ Final challenge (look down)

Before you tap the star, complete these tasks on paper.

Match each sub-problem to a suitable smaller task.

Match each item to a description from the list below.

Mark an item available / Check item availability / Add a member record

Item	Matching description
Members	
Loans	
Returns	

Why test each part before combining them?

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Your score

Copy it from the score screen as soon as it appears.

_____ / 22

Key question

How could a team share the work of building one program?

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Review and improve

Use review mode. Explain one correction and why the correct answer fits.

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Exam practice

Near the end of the lesson, close the experience and answer from memory.

a) Define decomposition. [2]

b) Draw a structure diagram for a club system with members, loans and returns. Decompose loans into two tasks. [4]

c) Explain one advantage of decomposition for a team. [2]

How confident am I?

Objective	Not yet	Getting there	Confident
Decompose a problem into sub-problems.			
Interpret and create structure diagrams.			
Explain benefits of decomposition.			